

Statistical Inference

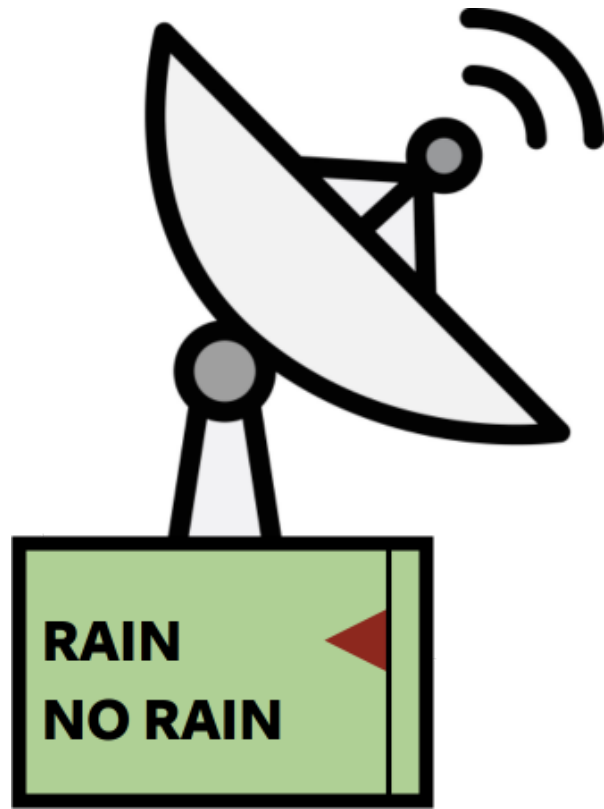
Drawing conclusions about a population

Statistics lets us learn about a larger population from a smaller sample.

How can we draw formal, rigorous conclusions?

- Hypothesis testing
- Confidence intervals

The Rain-O-Matic is a miraculous device that predicts whether there will be rain the next day.



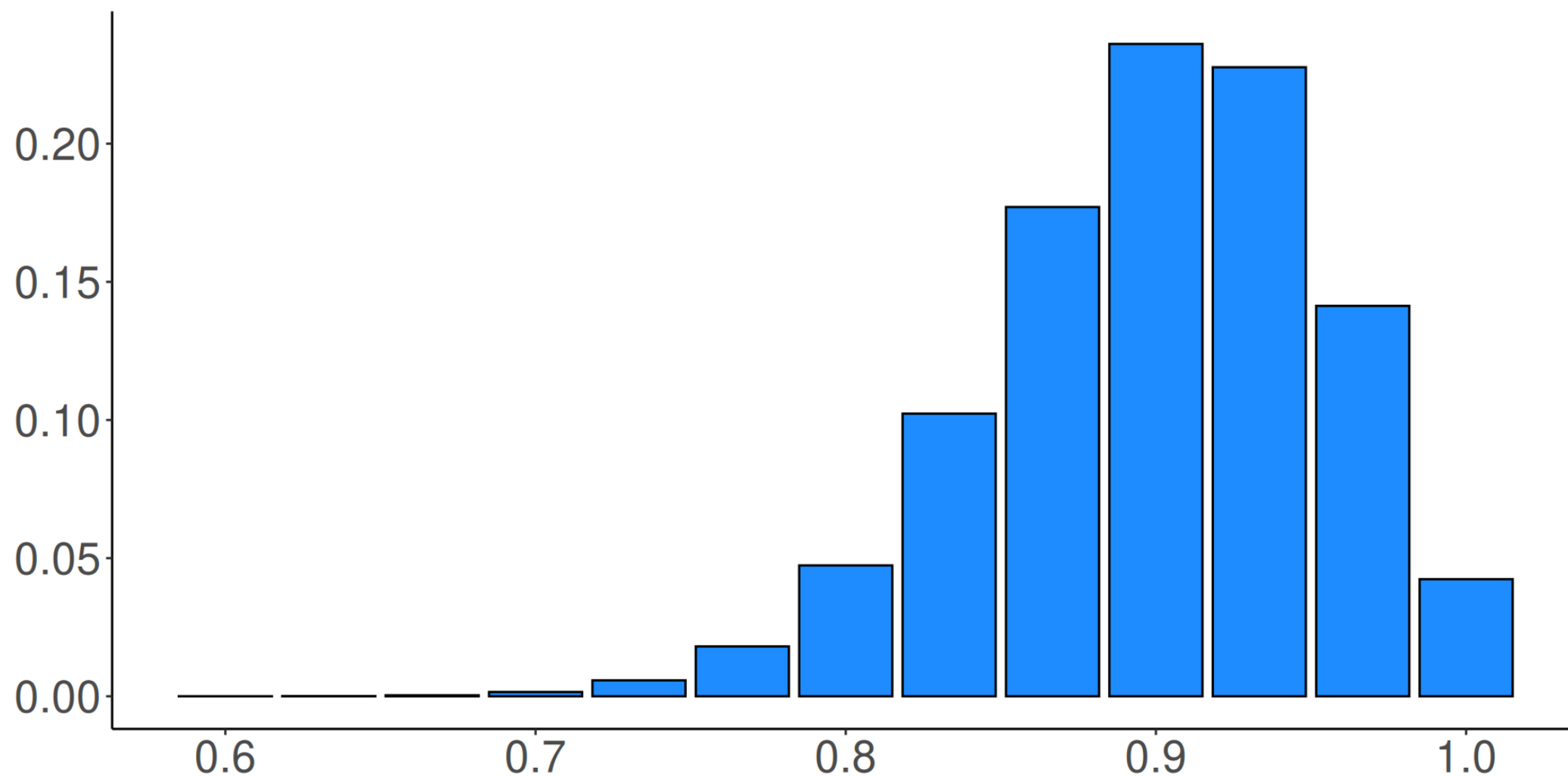
The device is advertised to work with “90% effectiveness”.

We test it out for 30 days and it works for 24 days. Is this noteworthy?

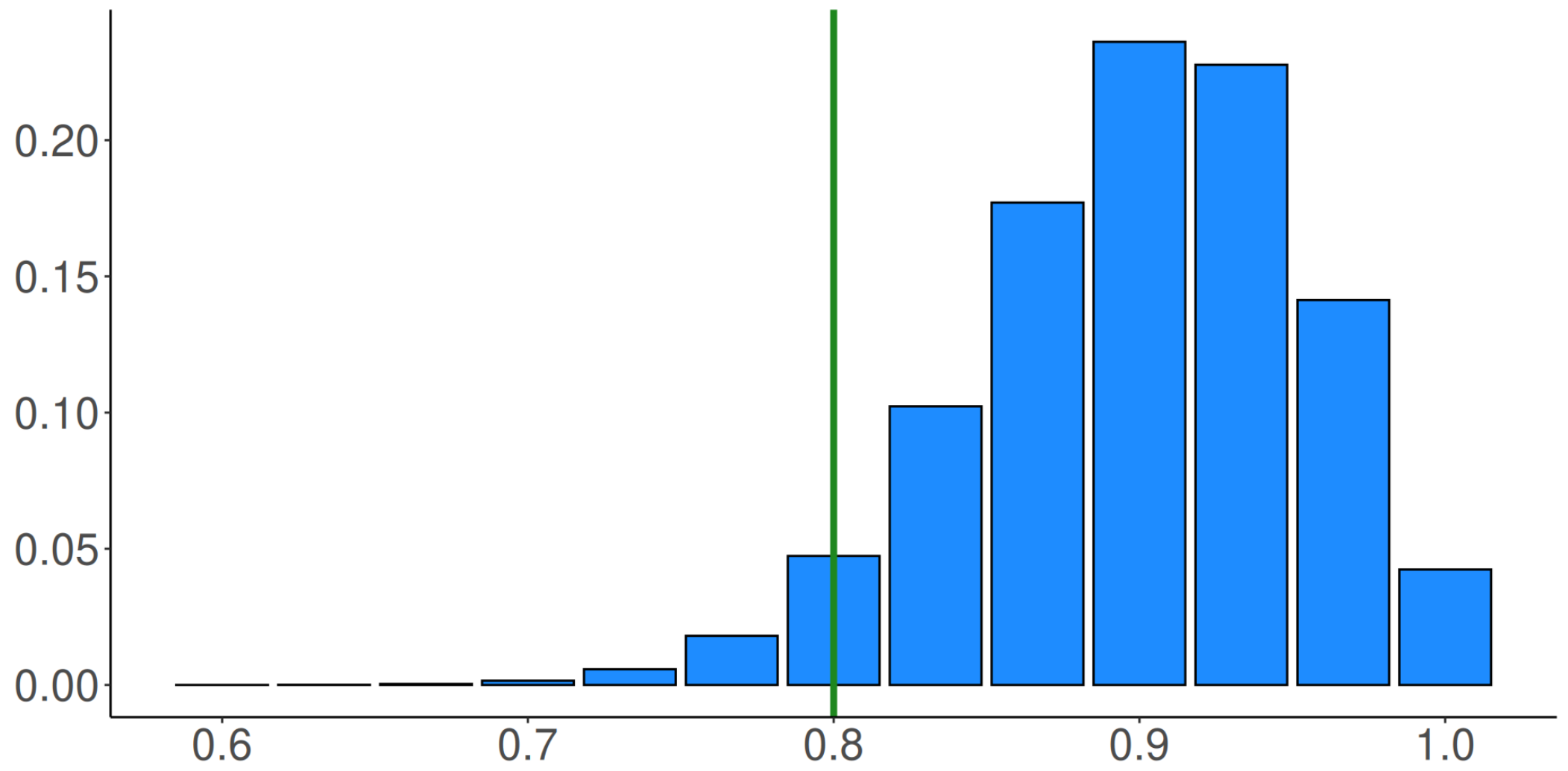
Let's consider what might happen if the advertisement is correct.

We already know how to do this! Set up a population that represents a 90% success rate, and take 30 draws.

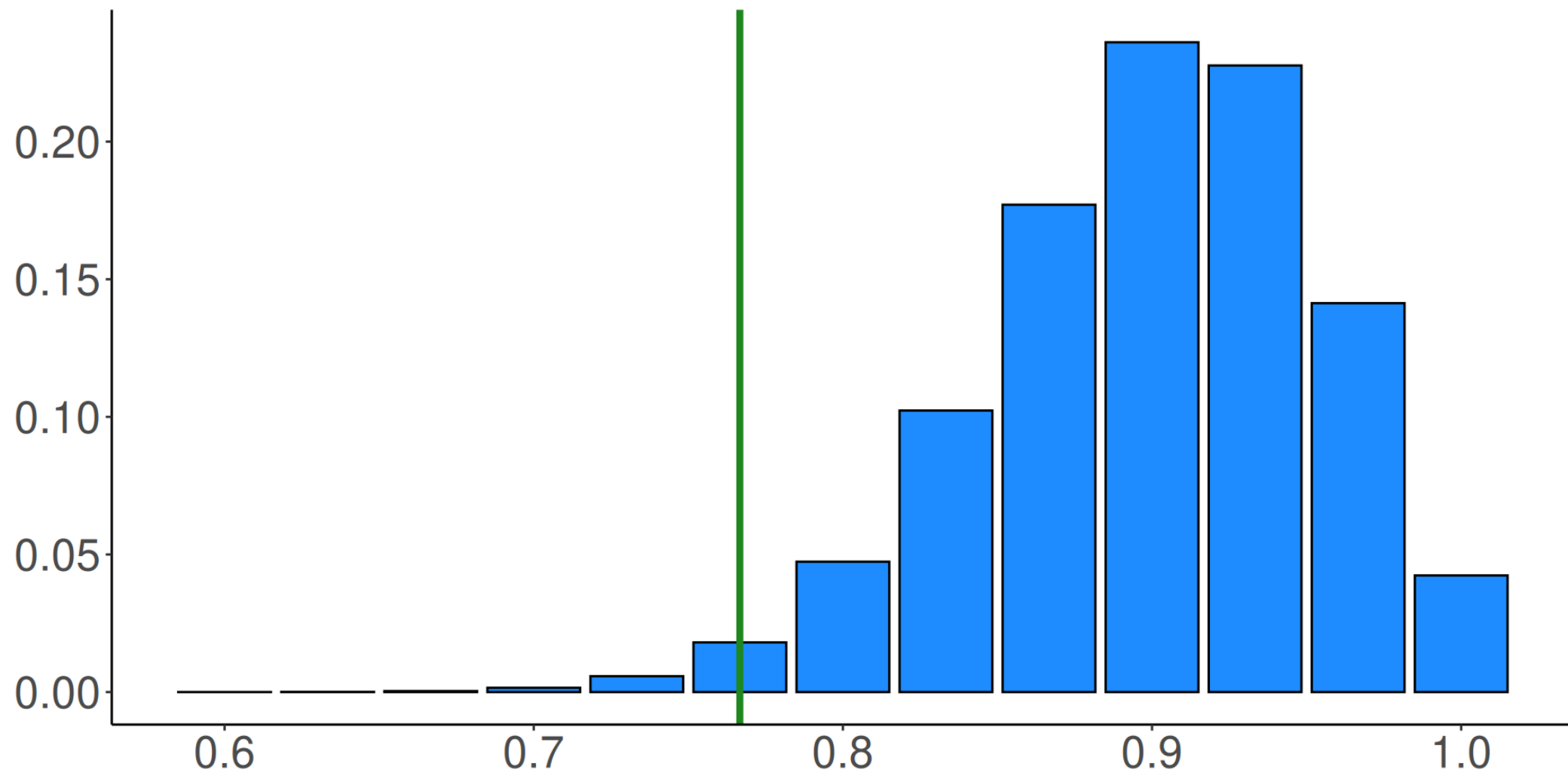
Sampling distribution, if the advertising is correct:



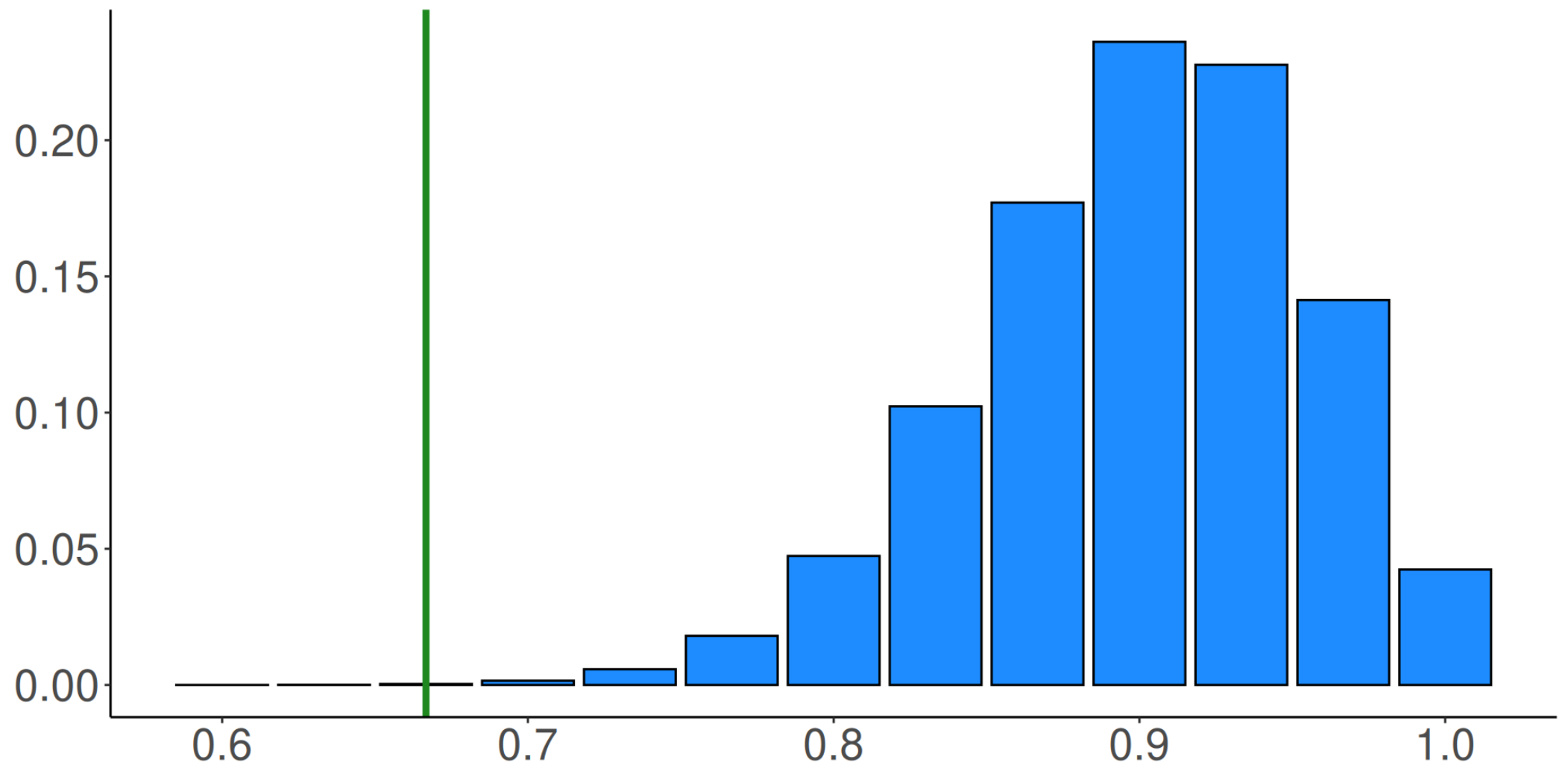
We observed 24/30, which is 0.8.



What if we observed 23/30 instead?



20/30...

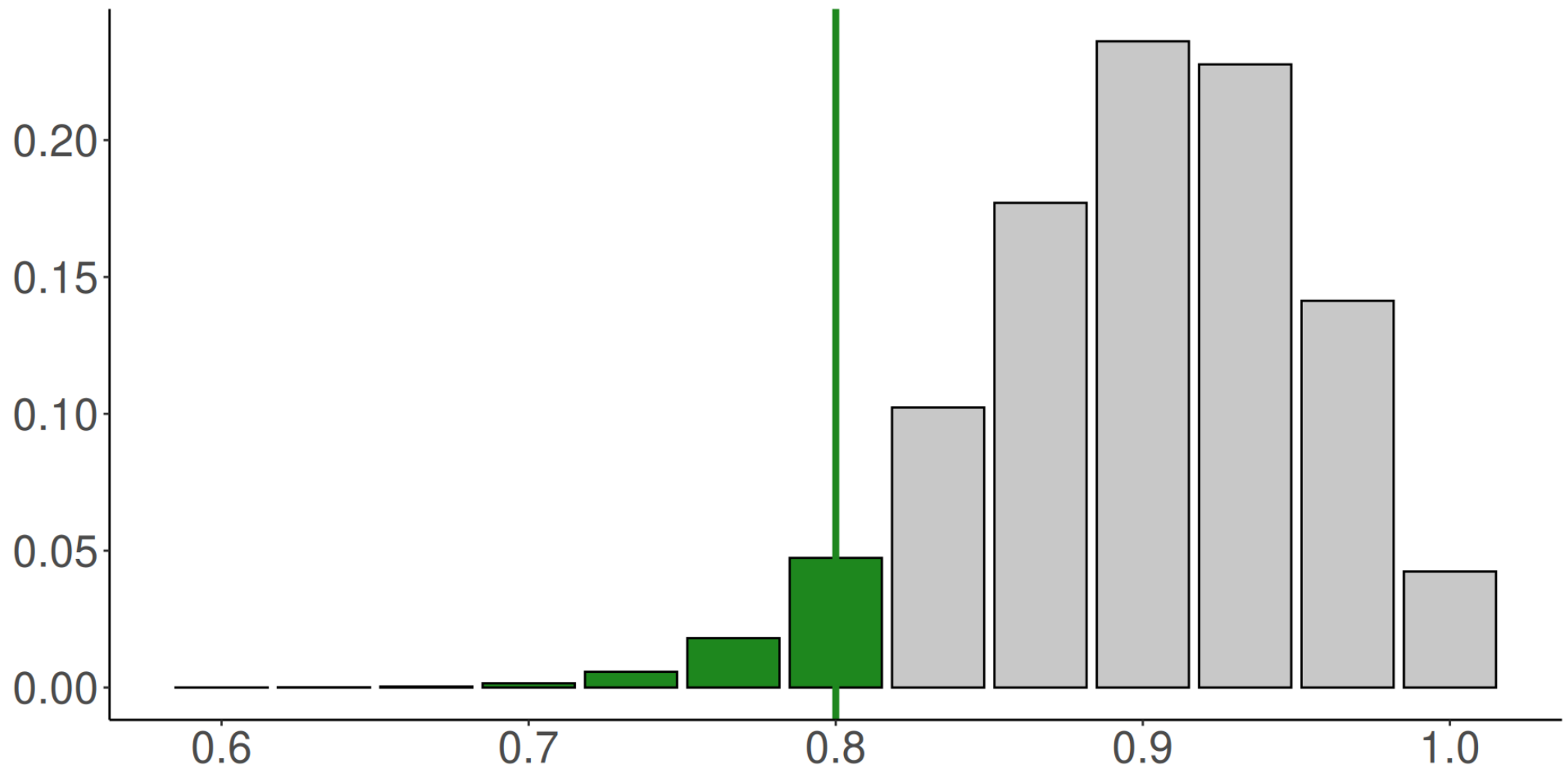


An observed value “outside” the histogram is less consistent with the advertisement.

How many histogram values are less than or equal to our observed value? This gives us a measure of “surprise”.

The advertisement is the **null hypothesis** and the measure of surprise is the **p-value**.

P-value for an observed 24/30 days correct:



Formally, we would write the null and alternative

$$H_0 : p = 0.9 \quad \text{versus} \quad H_A : p < 0.9$$

where p is the probability of a correct prediction.

A smaller p-value is more surprising (more evidence against H_0 , more evidence for H_A .)



In a sample of 300 births near power plants, 52% were girls. Is this noteworthy?

- Write a null and alternative hypothesis, assuming boy/girl births are supposed to be equally likely.
- Resample 10000 times from your null hypothesis to find the null distribution.
- Calculate a p-value based on the finding of 52%.



The p-value is a degree of evidence. How do we decide if we have *enough* evidence against the null?

We pick some kind of threshold, α . This is a small number from 0-1, conventionally 0.05.

- p-value < 0.05 : Reject H_0
- p-value ≥ 0.05 : Fail to reject H_0

In the births example, we get a p-value around 0.27, which is not enough evidence to reject.

This does not *prove* the null hypothesis! We just can't say for sure that it's wrong.

p-value: The probability of a result as or more extreme than our data, *if* the null were true.

The births example is *one-sided* since we're only looking at the case where girl births are more likely.

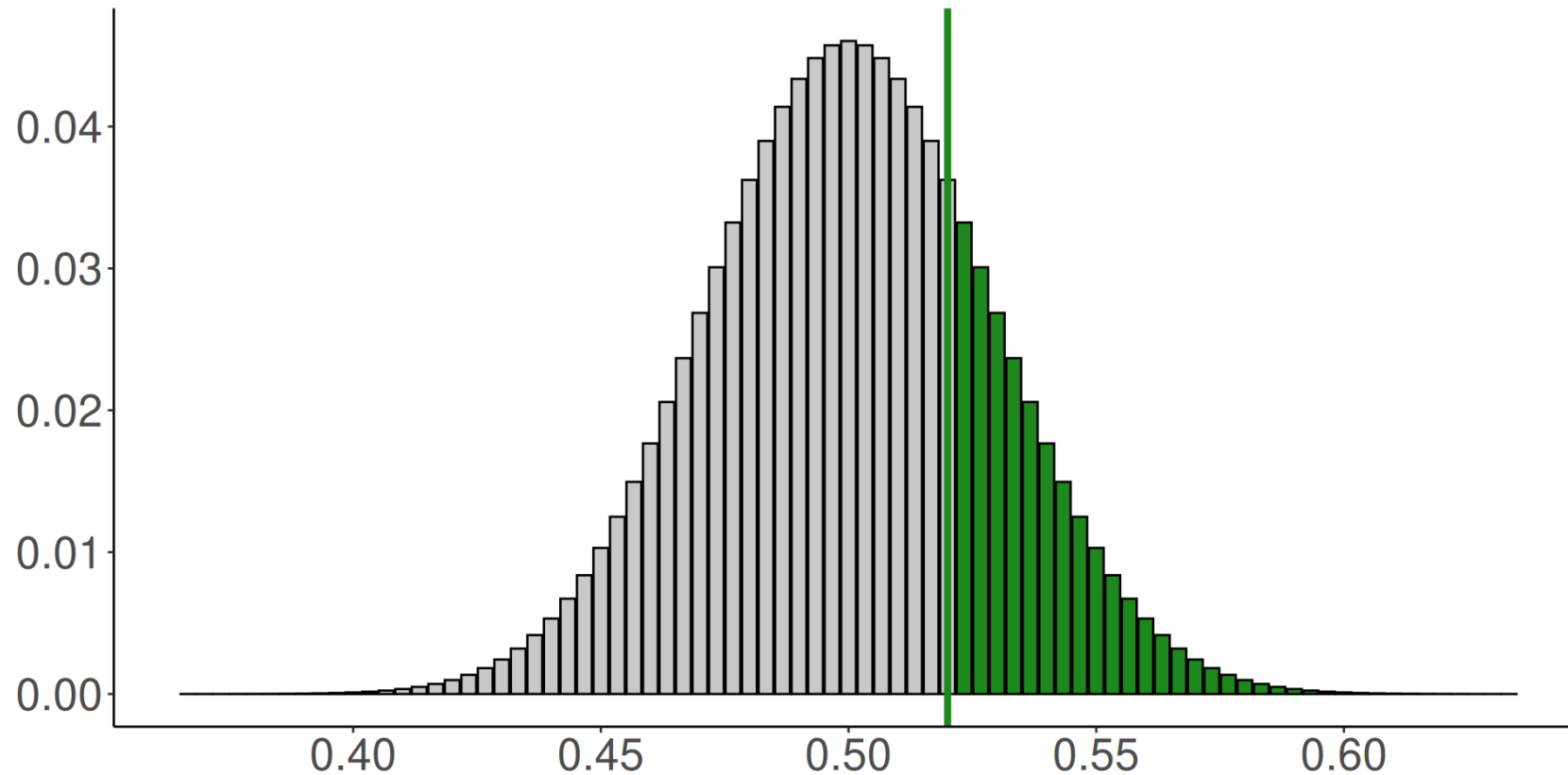
However, a 55/45 split in favor of girls seems like it should be just as noteworthy as a 45/55 split in favor of boys.

Idea: Consider *both* directions apart from 50-50 as interesting.

$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p \neq 0.5$$

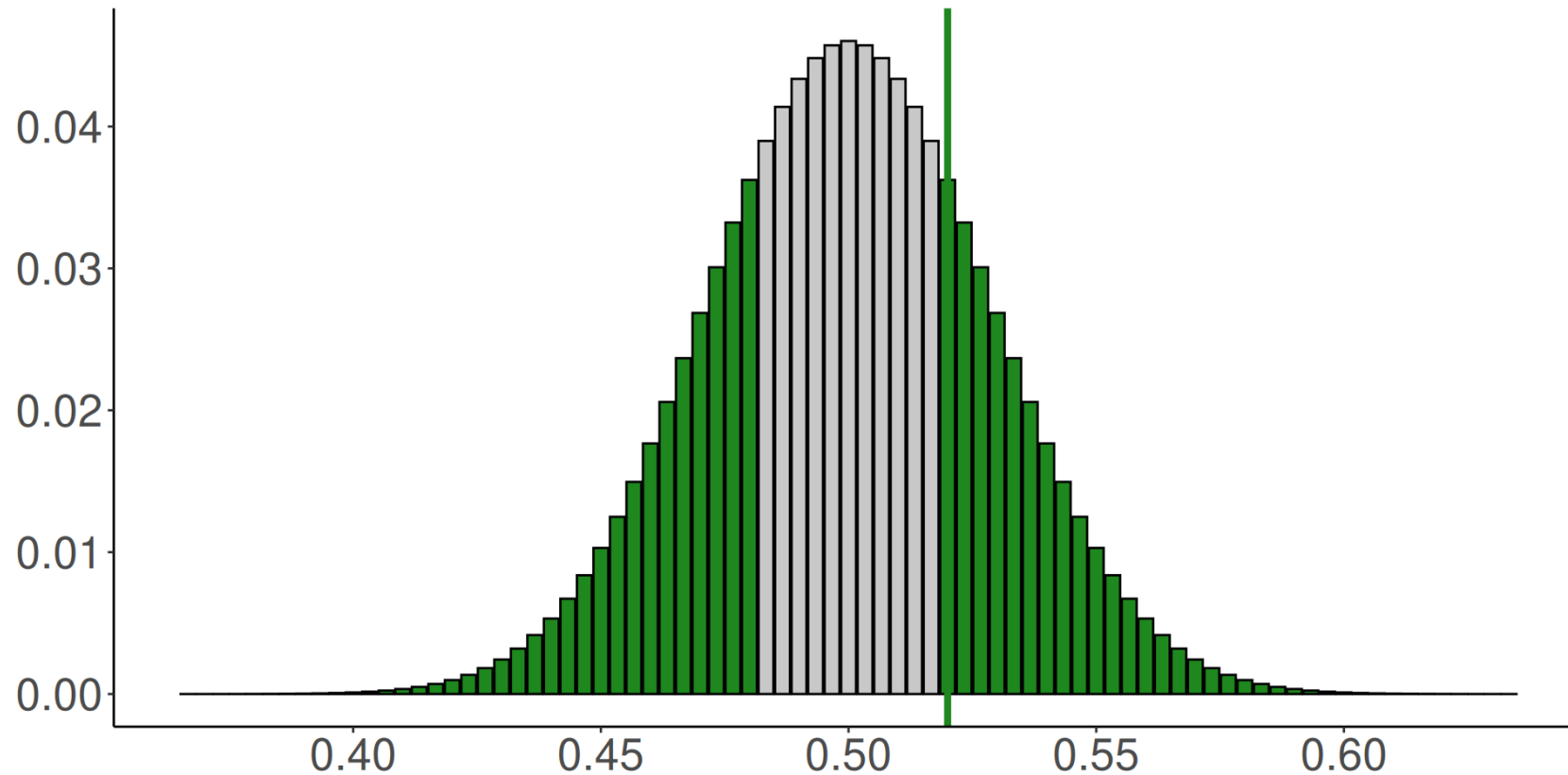
$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p > 0.5$$

One-sided pval = 0.27



$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p \neq 0.5$$

Two-sided pval = 0.54



When should we do a one- versus a two-sided test?

One-sided tests can only detect an effect in a single direction.

Cautious approach: always default to doing a two-sided test, unless the effect is only *possible* in one direction.



How should we choose our threshold α ?

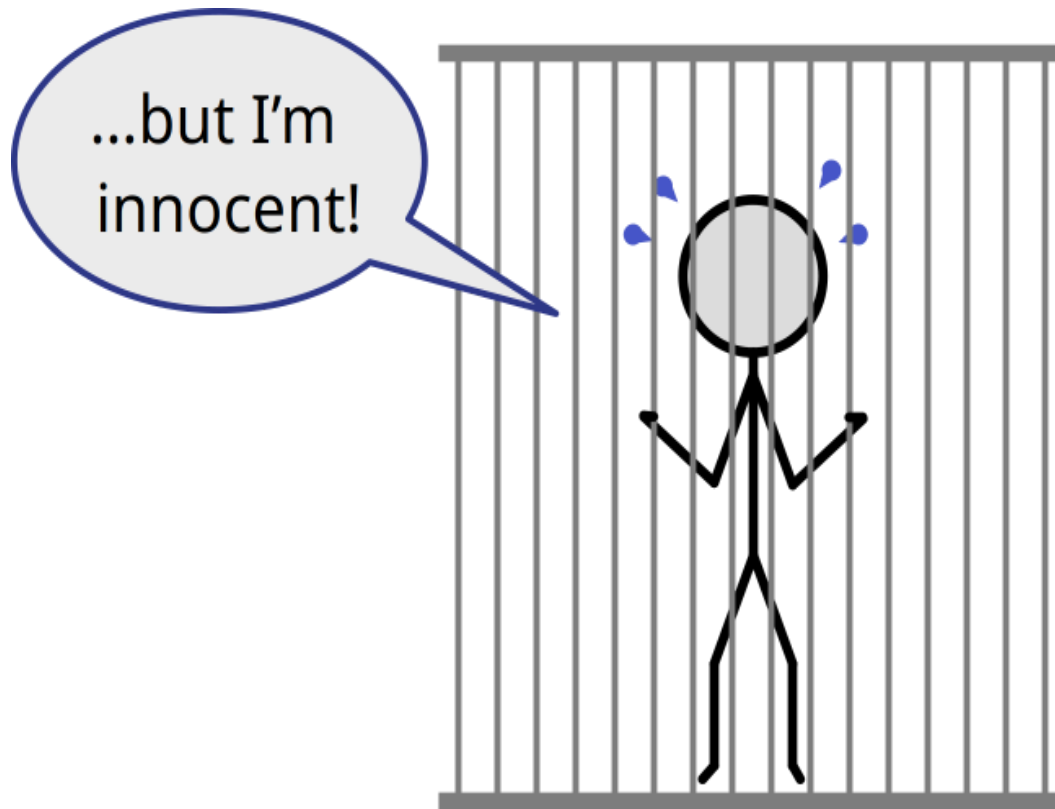
$\alpha = 0.05$ is an old convention. Some prefer $\alpha = 0.01$ or $\alpha = 0.001$, which makes it harder to reject H_0 .

H_0 is either true or false in reality. We decide to reject or fail to reject it. This leaves four cases:

	Reject	Fail to reject
H_0 true		
H_0 false		

- Label each cell as either an “error” or “correct decision”.

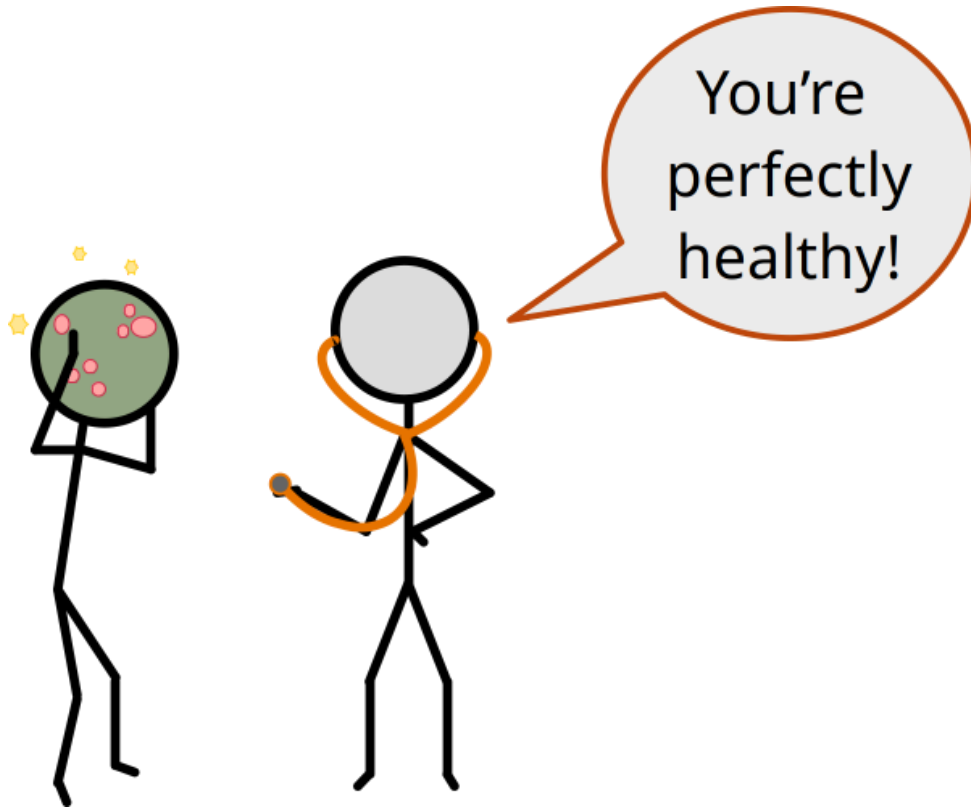




If the null is true, but we reject it, we have committed a **Type I** or **false positive** error.

$$\alpha = \text{Probability}(\text{Reject } H_0 \mid H_0 \text{ true})$$

We control this directly with α !



If the null is false, but we failed to it, we have committed a **Type II** or **false negative** error.

$$\beta = \text{Probability}(\text{Fail to reject } H_0 \mid H_0 \text{ false})$$

When α is lower, β is larger, and vice versa.

Type I

Type II

α

β

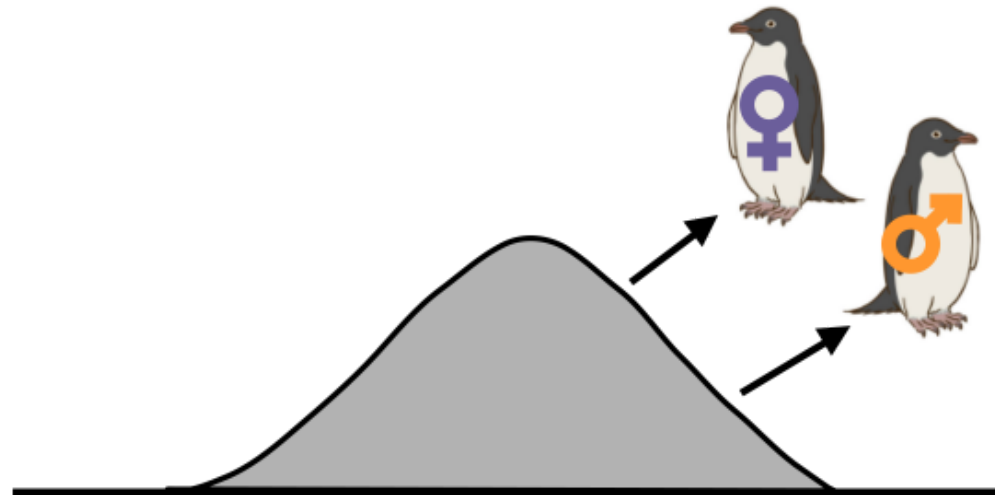
Reject but H_0 true

Fail to reject but H_0 false

False positive

False negative

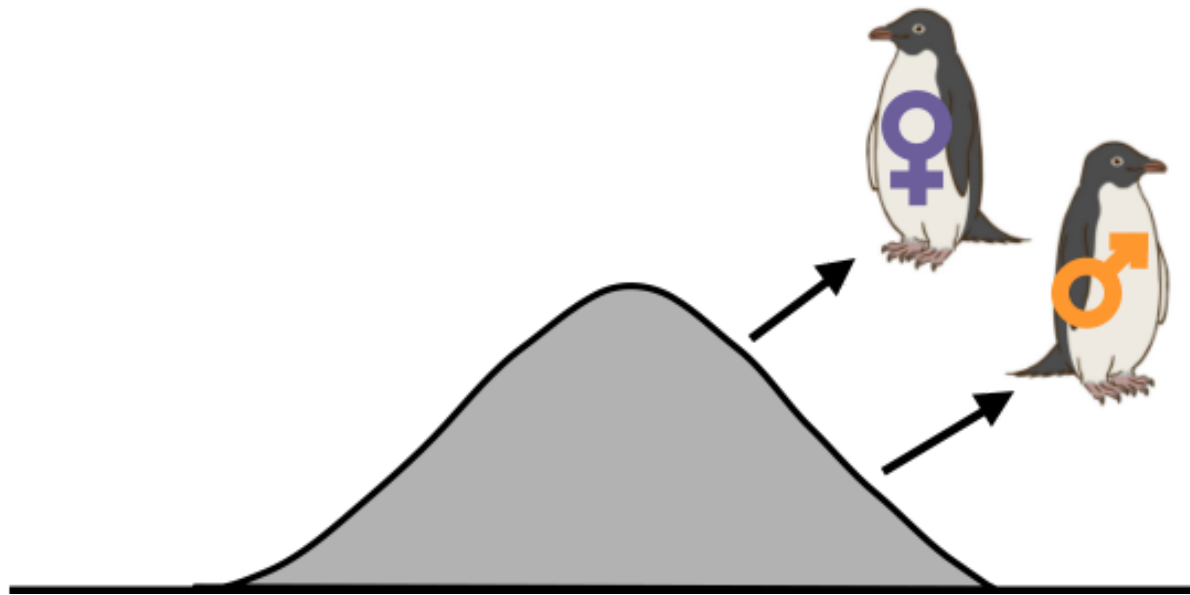
Recall the two-sample penguin body mass simulation experiment from last section.



The “big box” sampling approach implies a specific null hypothesis.

$H_0 : Med_F = Med_M$ versus $Med_F \neq Med_M$

$H_0 : Med_F - Med_M = 0$ versus $Med_F - Med_M \neq 0$



The observed difference in medians is completely separate from the resampled differences. (Both -600 and 600)!

This means our two-sided p-value is less than $\frac{1}{10000} = 0.0001$, a highly significant result.

We have enough evidence to reject H_0 .

In hypothesis testing, we start with an initial assumption about the population.

In **estimation**, we use our data to make a guess.

An **interval estimate** is a range of guesses, as opposed to a **point estimate**. Consider the average flipper length of Adelie penguins.



The high today will be between
-40 °F and 110 °F

The point estimate is
190.1. How wide
should our interval
be?



The high today will be between
66.3 °F and 66.5 °F

Let's incorporate α ,
our “willingness for
error”.

If $\alpha = 0.01$ (1% significance), let's consider the 99 “middle” values of our resampled means.

M_{lower} = 0.5th percentile, 50th value out of 10000

M_{upper} = 99.5th percentile, 9950th value out of 10000

(M refers to whatever we're studying)

We aren't quite there!

$$(M_{lower}, M_{upper})$$

is a statement about the *resamples*.

We need to take the original M_{obs} (190.1) into account:

$$(2M_{obs} - M_{upper}, 2M_{obs} - M_{lower})$$

In this experiment, $M_{obs} = 190.1$, $M_{lower} = 188.7$, and $M_{upper} = 191.5$.

The interval estimate is

$$(2(190.1) - 191.5, 2(190.1) - 188.7) = (188.7, 191.5)$$

This is a **99% confidence interval**. We are 99% confident that the true mean flipper length lies in this range.

How does the confidence level affect the interval width?

- Using the same set of re-samples, build a 95%, 98%, and 99% confidence interval.
- What is α for each one? What are the lower and upper quantiles for each one?
- How does the width change with the confidence level?



This method works as long as the distribution of

$$(M_{true} - M_{obs})$$

is similar to the distribution of

$$(M_{obs} - M_{resample})$$

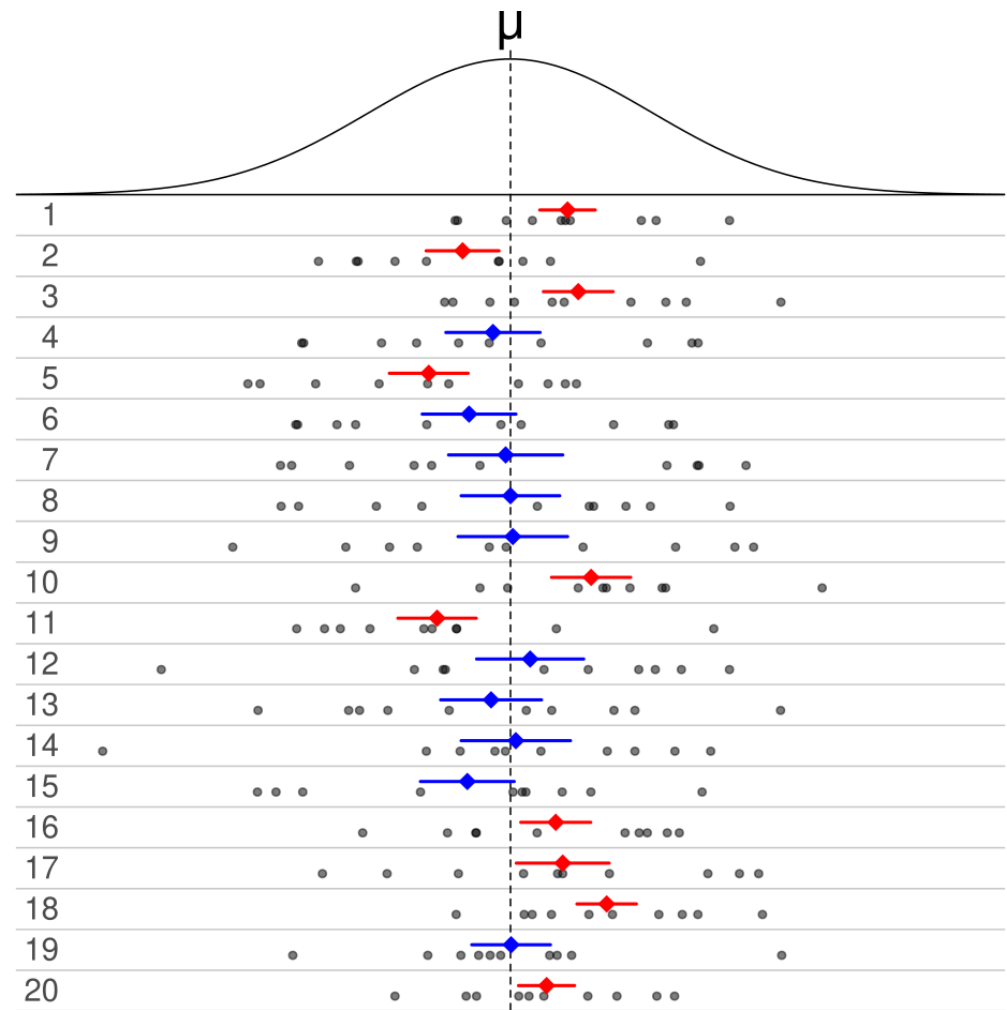
In other words, study the resamples from our data to learn how the data relates to the population.

Important note about interpretation! M_{true} is not random!

Once a CI is realized, it either covers M_{true} , or it does not.

“95%” or “99%” refers to the *success rate* of the procedure.

50% CIs, from Wikipedia:



Return to the example of studying the median Gentoo penguin flipper length.

- Build a 95% CI for the true median flipper length.
- Compare your CI to the assertion that the true median is 215.



Looking ahead

In [chapter 8](#), we'll learn some basic probability fundamentals for fitting statistical models. Consider the distribution of rolls of a weighted, un-fair six sided die called X :

x	1	2	3	4	5	6
$P(X = x)$	0.1	0.1	0.1	0.2	0.2	0.3

(X is also called a **random variable**). What is the average result of rolling die X ? Is this greater than 3.5, which is the average result of a fair six-sided die?